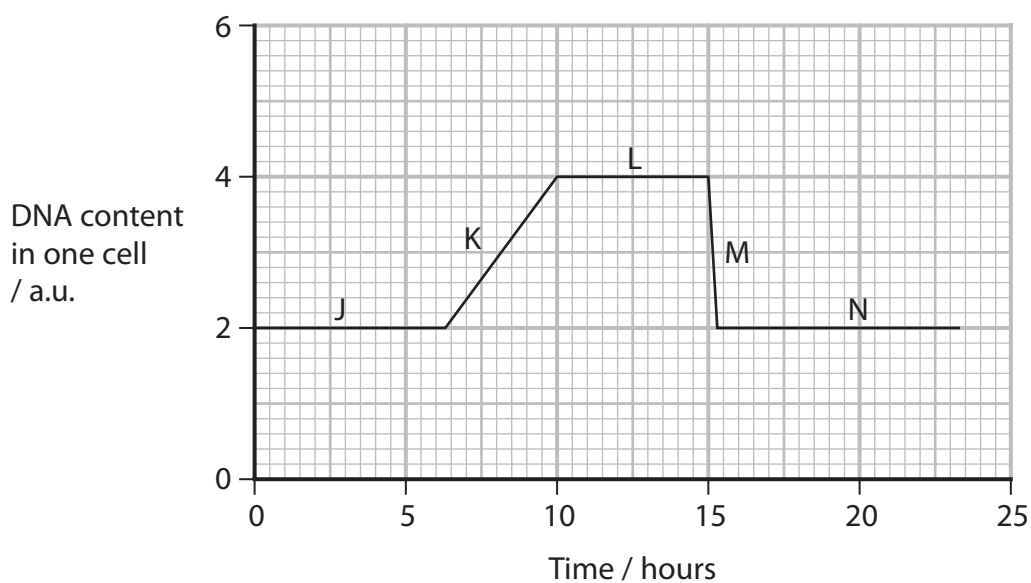


- 4 The graph shows the changes in the DNA content of one cell during one cell cycle.



- (a) (i) In which part of the graph is the DNA being replicated?

(1)

- ☐ **A** J
- ☐ **B** K
- ☐ **C** L
- ☐ **D** M

- (ii) In which part of the graph would the condensation of chromosomes be completed?

(1)

- ☐ **A** J
- ☐ **B** L
- ☐ **C** M
- ☐ **D** N



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(b) Centrioles are involved in spindle formation.

(i) Name the stage of mitosis when spindle fibres begin to form.

(1)

(ii) Explain the role of the spindle in mitosis.

(2)

(iii) Draw a plant cell undergoing cell division, after mitosis has just finished.

(2)



(c) The photograph shows a section through a root tip, as seen using a microscope.



(Source: © agefotostock / Alamy Stock Photo)

Describe how to determine the mitotic index of this tissue.

(2)

(Total for Question 4 = 9 marks)

